I Correlation

Investigation

At a junior tournament, some young athletes each throw a discus. The *age* and *distance thrown* are recorded for each athlete.

Athlete	A	B	C	D	E	F	G	H	I	J	K	L
Age (years)	12	16	16	18	13	19	11	10	20	17	15	13
Distance thrown (m)	20	35	23	38	27	47	18	15	50	33	22	20

Things to think about :

a Do you think the distance an athlete can throw is related to the person's age ?

b What happens to the distance thrown as the age of the athlete increases ?

c How could you graph the data to more clearly see the relationship between the variables ?

d How can we *measure* the relationship between the variables ?

Each athlete has had *two* variables (*age* and *distance thrown*) recorded about them. This type of data is called **bivariate data**. We study it to understand the relationship between the two variables.

We expect the distance thrown will depend on the athlete's age, so age is the independent variable and distance thrown is the dependent variable.

Remark. The independent and dependent variables are sometimes called the explanatory and response variables respectively.

Definition

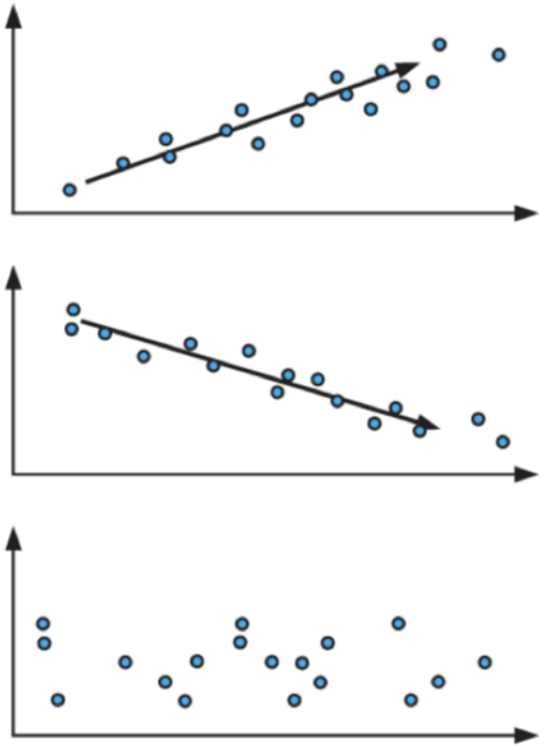
Correlation refers to the relationship or association between two numerical variables.

Definition

For a generally upward trend, we say that the correlation is **positive**. An increase in the independent variable generally results in an increase in the dependent variable.

For a generally downward trend, we say that the correlation is **negative**. An increase in the independent variable generally results in a decrease in the dependent variable.

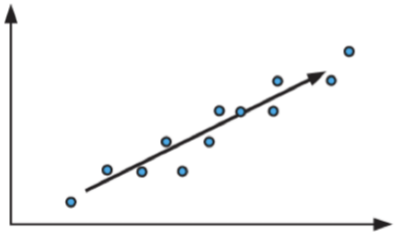
For randomly scattered points, with no upward or downward trend, we say there is **no correlation**.



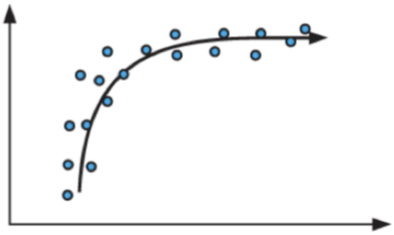
Definition

When a trend exists, if the points approximately form a straight line, we say the trend is **linear**.

These points are roughly linear.

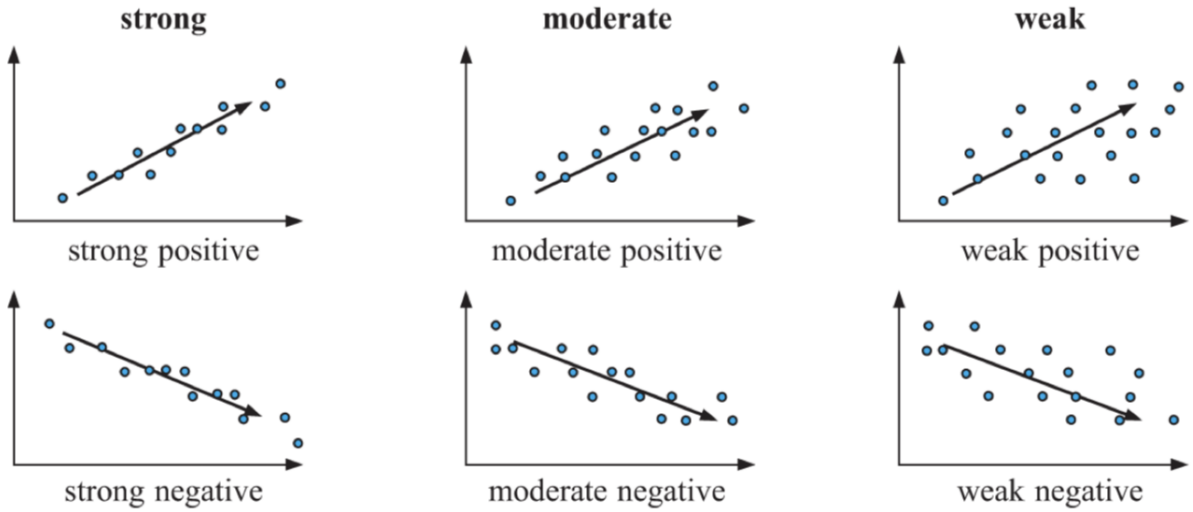


These points do not follow a linear trend.



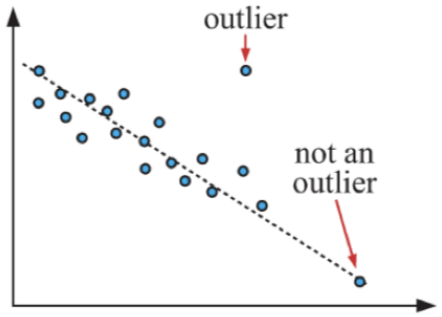
Definition

To describe how closely the data follows a pattern or trend, we talk about the **strength** of correlation. It is usually described as either **strong**, **moderate**, or **weak**.



Definition

Outliers are isolated points which do not follow the trend formed by the main body of data.



Remark. If an outlier is the result of a recording or graphing error, it should be discarded.
However, if the outlier is a genuine piece of data, it should be kept.

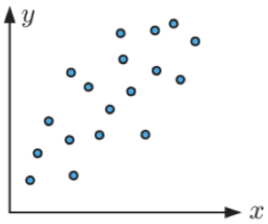
Remark. Correlation between two variables does not necessarily mean that one variable *causes* the other.

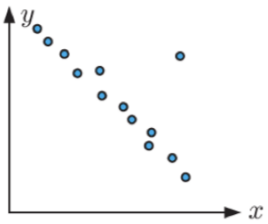
Definition

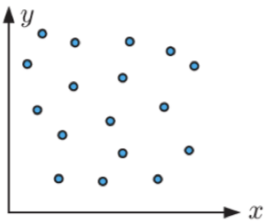
If a change in one variable *causes* a change in the other variable then we say that a **causal relationship** exists between them. In these cases, we can say that the independent variable *explains* the dependent variable. It may be more natural to use the terminology **explanatory variable** and **response variable**.

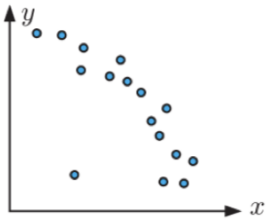
EXERCISE 1

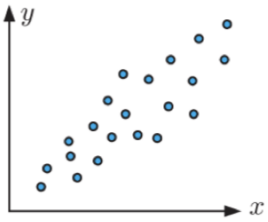
For each scatter diagram, describe the relationship between the variables. Consider the direction, linearity, and strength of the relationship, as well as the presence of any outliers.

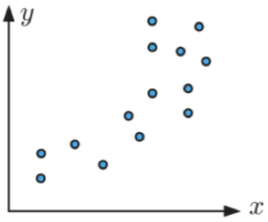
a 

b 

c 

d 

e 

f 

EXERCISE 2

The scores awarded by two judges at an ice skating competition are shown in the table.

Competitor	P	Q	R	S	T	U	V	W	X	Y
Judge A	5	6.5	8	9	4	2.5	7	5	6	3
Judge B	6	7	8.5	9	5	4	7.5	5	7	4.5

a Construct a scatter diagram for the data, with Judge A's scores on the horizontal axis and Judge B's scores on the vertical axis.

b Copy and complete the following comments about the scatter diagram:
There appears to be ....., ..., ... correlation between Judge A's scores and Judge B's scores. This means that as Judge A's scores increase, Judge B's scores .....

c Would it be reasonable to conclude that an increase in Judge A's scores *causes* an increase in Judge B's scores? Explain your answer.

EXERCISE 3

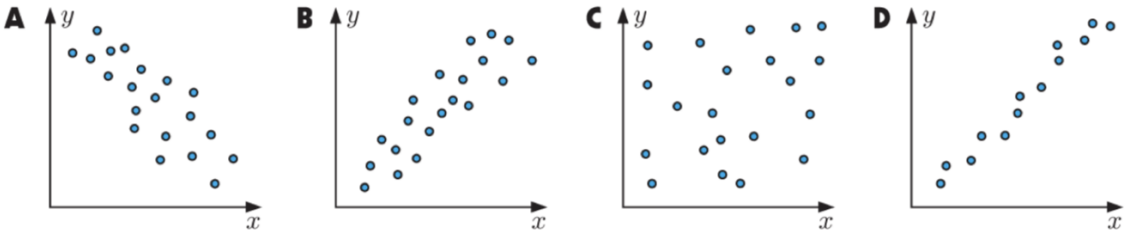
Choose the scatter diagram which would best illustrate the relationship between the variables x and y.

a $x =$ the number of apples bought by customers, $y =$ the total cost of apples bought

b $x =$ the number of pushups a student can perform in one minute,
$y =$ the time taken for the student to run 100 metres

c $x =$ the height of a person, $y =$ the weight of the person

d $x =$ the distance a student travels to school, $y =$ the height of the student's uncle



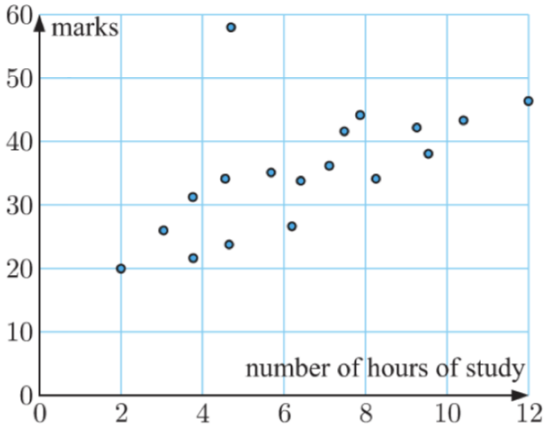
EXERCISE 4

The scatter diagram shows the marks obtained by students in a test out of 50 marks, plotted against the number of hours each student studied for the test.

a Describe the correlation between the variables.

b How should the outlier be treated? Explain your answer.

c Do you think there is a causal relationship between the variables? Explain your answer.



II Pearson's Correlation Coefficient

Since the method is *subjective* and relies on the observer's opinion, it is important to get a more precise measure of the strength of linear correlation between the variables. Hence the use of **Pearson's product-moment correlation coefficient** r.

Property

For a set of n data given as ordered pairs $(x_1,y_1), (x_2,y_2), (x_3,y_3), \ldots, (x_n,y_n)$,

$$r = \frac{\sum(x-\overline{x})(y-\overline{y})}{\sqrt{\sum(x-\overline{x})^2 \sum(y-\overline{y})^2}}$$

where $\overline{x}$ and $\overline{y}$ are the means of the x and y data respectively, and $\sum$ means the sum over all the data values.

Property

- The values of r range from -1 to $+1$.
- The **sign** of r indicates the **direction** of the correlation.
  - A positive value for r indicates the variables are **positively correlated**. An increase in one variable results in an increase in the other.
  - A negative value for r indicates the variables are **negatively correlated**. An increase in one variable results in a decrease in the other.
  - If $r = 0$ then there is **no correlation** between the variables.
- The **size** of r indicates the **strength** of the correlation.
  - A value of r close to $+1$ or -1 indicates strong correlation between the variables.
  - A value of r close to zero indicates weak correlation between the variables.

Method.

Positive correlation		Negative correlation	
$r = 1$	perfect positive correlation	$r = -1$	perfect negative correlation
$0.95 \leqslant r < 1$	very strong positive correlation	$-1 < r \leqslant -0.95$	very strong negative correlation
$0.87 \leqslant r < 0.95$	strong positive correlation	$-0.95 < r \leqslant -0.87$	strong negative correlation
$0.7 \leqslant r < 0.87$	moderate positive correlation	$-0.87 < r \leqslant -0.7$	moderate negative correlation
$0.5 \leqslant r < 0.7$	weak positive correlation	$-0.7 < r \leqslant -0.5$	weak negative correlation
$0 < r < 0.5$	very weak positive correlation	$-0.5 < r < 0$	very weak negative correlation

Example

The botanical gardens have been trying a new chemical to control the number of beetles infesting their plants. The results of one of their tests are shown in the table.

Sample	*Quantity of chemical (g)*	*Number of surviving beetles*
A	2	11
B	5	6
C	6	4
D	3	6
E	9	3

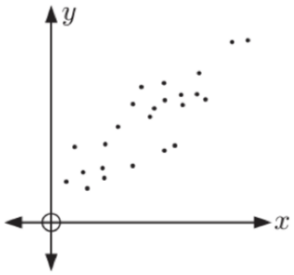
a Draw a scatter diagram for the data.

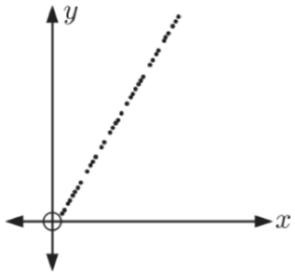
b Determine the correlation coefficient r.

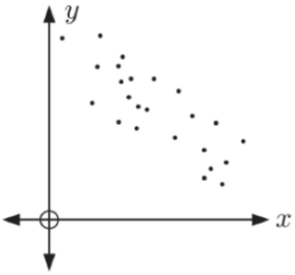
c Describe the correlation between the *quantity of chemical* and the *number of surviving beetles*.

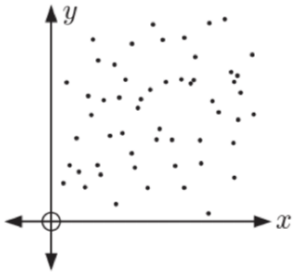
EXERCISE 5

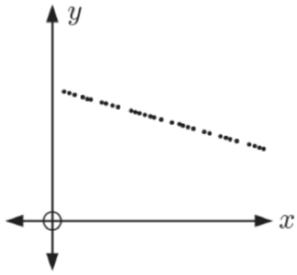
Match each scatter diagram with the correct value of r.

a 

b 

c 

d 

e 

A $r = 1$ **B** $r = 0.6$ **C** $r = 0$ **D** $r = -0.7$ **E** $r = -1$

EXERCISE 6

For each of the following data sets:

i Draw a scatter diagram for the data.
ii Calculate Pearson's product-moment correlation coefficient r.
iii Describe the linear correlation between x and y.

a

x	1	2	3	4	5	6
y	3	2	5	5	9	6

b

x	3	8	5	14	19	10	16
y	17	12	15	6	1	10	4

c

x	3	6	11	7	5	6	8	10	4
y	2	8	8	4	7	9	11	1	5

EXERCISE 7

A selection of students was asked how many phone calls and text messages they received the previous day. The results are shown alongside.

Student	A	B	C	D	E	F	G	H
Phone calls received	4	7	1	0	3	2	2	4
Text messages received	6	9	2	2	5	8	4	7

a Draw a scatter diagram for the data.
b Calculate r.
c Describe the linear correlation between *phone calls received* and *text messages received*.
d Give a reason why this correlation may occur.

EXERCISE 8

This table shows the number of supermarkets in 10 towns, and the number of car accidents that have occurred in these towns in the last month.

Number of supermarkets	5	8	12	7	6	2	15	10	7	3
Number of car accidents	10	13	27	19	10	6	40	30	22	37

a Draw a scatter diagram for the data.

b Calculate r.

c Identify the outlier in the data.

d It was found that the outlier was due to an error in the data collection process.

i Recalculate r with the outlier removed.

ii Describe the relationship between the variables.

iii Discuss the effect of removing the outlier on the value of r.

e Do you think there is a causal relationship between the variables? Explain your answer.

III Line of Best Fit

If there is a sufficiently strong linear correlation between two variables, we can draw a **line of best fit** to illustrate their relationship. In general, it is only worth drawing a line of best fit if the correlation between the variables is strong. There is no fixed rule, but we suggest that a line of best fit is not appropriate if $|r| < 0.85$.

Method (Drawing a line of best fit by eye)**.**

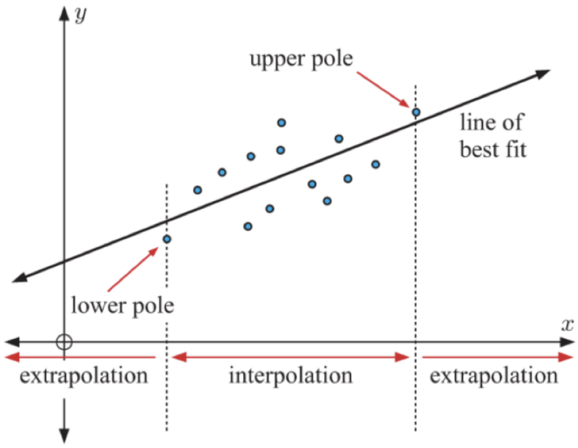
Step 1 : Calculate the mean of the x values $\overline{x}$, and the mean of the y values $\overline{y}$.

Step 2 : Mark the mean point $(\overline{x}, \overline{y})$ on the scatter diagram.

Step 3 : Draw a line through the mean point which fits the trend of the data, and so that about the same number of data points are above the line as below it.

Definition

Consider the data in the scatter diagram alongside. The data with the highest and lowest values are called the **poles**.



We can use this line to predict the value of one variable for a given value of the other.

- If we predict a y value for an x value in between the poles, we say we are **interpolating** in between the poles.
- If we predict a y value for an x value outside the poles, we say we are **extrapolating** outside the poles.

Example

On a hot day, six cars were left in the sun in a car park. The length of time each car was left in the sun was recorded, as well as the temperature inside the car at the end of the period.

Car	A	B	C	D	E	F
Time (x minutes)	50	5	25	40	15	45
Temperature ($y\,^\circ C$)	47	28	36	42	34	41

a Calculate $\overline{x}$ and $\overline{y}$.

b Draw a scatter diagram for the data.

c Locate the mean point $(\overline{x}, \overline{y})$ on the scatter diagram, then draw a line of best fit through this point.

d Predict the temperature of a car which has been left in the sun for 35 minutes.

e Predict how long it would take for a car's temperature to reach 55°C.

f Comment on the reliability of your predictions in **d** and **e**.

EXERCISE 9

Fifteen students were weighed and their pulse rates were measured:

Weight (x kg)	46	37	32	57	47	64	42	30	52	56	65	43	36	28	40
Pulse rate (y beats per min)	65	59	54	74	69	87	61	59	70	69	75	60	56	53	58

a Draw a scatter diagram for the data. b Calculate r.

c Describe the relationship between *weight* and *pulse rate*.

d Calculate the mean point $(\overline{x}, \overline{y})$.

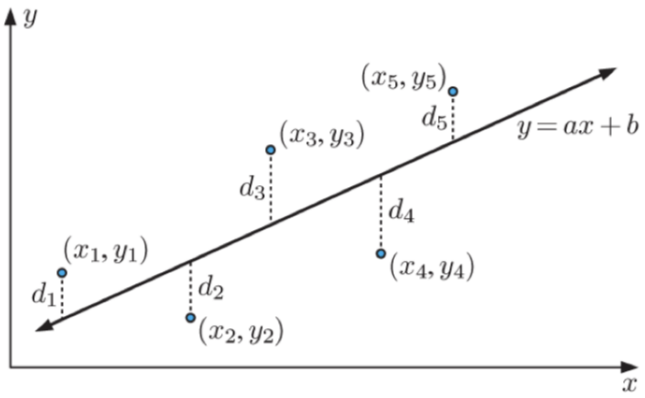
e Locate the mean point on the scatter diagram, then use it in drawing a line of best fit.

f Estimate the pulse rate of a 50 kg student. Comment on the reliability of your estimate.

IV Least Squares Regression

The problem with drawing a line of best fit by eye is that the line drawn will vary from one person to another. For consistency, we use a method known as **linear regression** to find the equation of the line which best fits the data. The most common method is the method of "least squares".

In least squares linear regression, we minimise the sum of the squares of the *vertical* distances between each data point and the **regression line**.



Property

The required values of a and b are

$$a = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})}{\sum(x_i - \overline{x})^2} \quad \text{and} \quad b = \overline{y} - a\overline{x}.$$

Remark. In this course you will not be required to find the equation of the least squares regression line by hand.

Example

The annual income and average weekly grocery bill for a selection of families is shown below :

Income (x thousand pounds)	55	36	25	47	60	64	42	50
Grocery bill (y pounds)	120	90	60	160	190	250	110	150

a Construct a scatter diagram to illustrate the data.

b Find the equation of the regression line. State and interpret its gradient.

c Estimate the weekly grocery bill for a family with an annual income of £95 000.

d Estimate the annual income of a family whose weekly grocery bill is £100.

e Comment on whether the estimates in **c** and **d** are likely to be reliable.

EXERCISE 10

Steve wanted to see whether there was any relationship between the temperature when he leaves for work in the morning, and the time it takes for him to get to work.
He collected data over a 14 day period:

Temperature (x °C)	25	19	23	27	32	35	29	27	21	18	16	17	28	34
Time (y minutes)	35	42	49	31	37	33	31	47	42	36	45	33	48	39

a Draw a scatter diagram for the data. b Calculate r.

c Describe the relationship between the variables.

d Is it reasonable to fit a linear model to this data? Explain your answer.

EXERCISE 11

The prices of petrol and the number of customers per hour for sixteen petrol stations are:

Petrol price (x cents per litre)	105.9	106.9	109.9	104.5	104.9	111.9	110.5	112.9
Number of customers (y)	45	42	25	48	43	15	19	10

Petrol price (x cents per litre)	107.5	108.0	104.9	102.9	110.9	106.9	105.5	109.5
Number of customers (y)	30	23	42	50	12	24	32	17

a Calculate Pearson's product-moment correlation coefficient for the data.

b Describe the relationship between the *petrol price* and the *number of customers*.

c Use technology to find the equation of the regression line.

d State and interpret the gradient of the regression line.

e Estimate the number of customers per hour for a petrol station which sells petrol at 115.9 cents per litre.

f Estimate the petrol price at a petrol station which has 40 customers per hour.

g Comment on the reliability of your estimates in **e** and **f**.

EXERCISE 12

A group of children was asked the numbers of hours they spent exercising and watching television each week.

Exercise (x hours per week)	4	1	8	7	10	3	3	2
Television (y hours per week)	12	24	5	9	1	18	11	16

a Draw a scatter diagram for the data. **b** Calculate r.

c Describe the correlation between *time exercising* and *time watching television*.

d Find the equation of the regression line, and draw the line on your scatter diagram.

e State and interpret the gradient and y-intercept of the regression line.

f **i** One of the children in the group exercised for 7 hours each week. How much television does this child watch weekly?

ii Use the regression line to predict the amount of television watched each week by a child who exercises for 7 hours each week.

iii Compare your answers to **i** and **ii**.

EXERCISE 13

The yield of pumpkins on a farm depends on the quantity of fertiliser used.

Fertiliser (x g per m^2)	4	13	20	26	30	35	50
Yield (y kg)	1.8	2.9	3.8	4.2	4.7	5.7	4.4

a Draw a scatter diagram for the data, and identify the outlier.

b What effect do you think the outlier has on:

i the strength of correlation of the data ii the gradient of the regression line?

c Calculate the correlation coefficient:

i with the outlier included ii without the outlier.

d Calculate the equation of the regression line:

i with the outlier included ii without the outlier.

e If you wish to estimate the yield when 15 g per m^2 of fertiliser is used, which regression line from **d** should be used? Explain your answer.

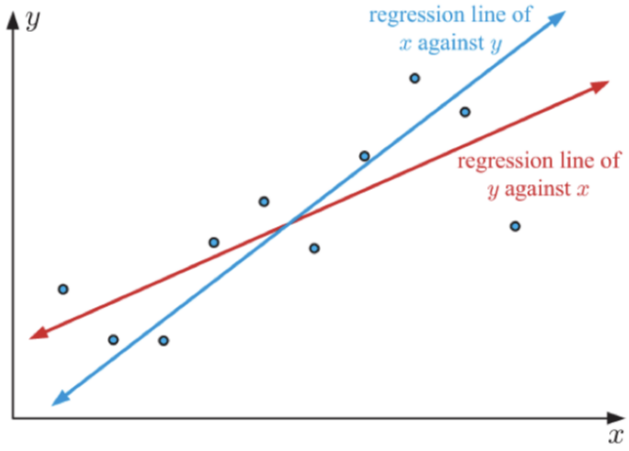
f Can you explain what may have caused the outlier? Do you think the outlier should be kept when analysing the data?

V The Regression Line of x on y

Example

When a student studies for a test, their time spent studying x explains their test score y. However, the test score will be more precisely measured than the amount of time spent studying.

At a breath testing station, police use a breathalyser to estimate the blood alcohol concentration (BAC) of drivers. If the result x is sufficiently high, the driver is required to take a blood test to establish their actual BAC. The blood test result y is a much more precise measurement.



In these scenarios, we consider the regression line of x against y. This means that we minimise the *horizontal* distances of points from the line, so all of the "error" is associated with the explanatory variable x.

Example

The data below shows the quantity of feed eaten and the number of eggs laid in a fortnight for 12 Leghorn chickens :

Feed (x kg)	0.57	0.86	0.49	1.37	0.91	0.50
Number of eggs (y)	4	6	3	9	6	3

Feed (x kg)	0.97	1.06	1.00	0.68	1.34	0.94
Number of eggs (y)	6	7	7	4	9	6

a Explain why it would be appropriate to use the regression line of x against y in this case.

b Find the regression line of x against y.

c Use the regression line to estimate :

i the quantity of feed required for a Leghorn chicken to lay a dozen eggs

ii the number of eggs laid in a fortnight by a hen that ate 1.2 kg of feed.

EXERCISE 14

The table below shows how far a group of students live from school, and how long it takes them to travel there each day.

Distance from school (x km)	7.2	4.5	13	1.3	9.9	12.2	19.6	6.1	23.1
Time to travel to school (y min)	17	13	29	2	25	27	41	15	53

a Draw a scatter diagram of the data.

b Which regression line should be used to model the relationship between the variables? Explain your answer.

c Use an appropriate regression line to estimate the travel time of a student who lives 15 km from school.

d Comment on the reliability of your estimate.

EXERCISE 15

Eight students swim 200 m breaststroke. Their times y in seconds, and arm lengths x in cm, are shown in the table below:

Length of arm (x cm)	78	73	71	68	76	72	63	69
Breaststroke (y seconds)	123.1	123.7	127.3	132.0	120.8	125.0	140.9	129.0

a Draw a scatter diagram for the data.

b Find the equation of the regression line of:

i y against x **ii** x against y.

c Plot both regression lines on your scatter diagram. What do you notice?